CE541E08 — Introduction to Python and MATLAB Programming

Unit 1 · Day 6 — Operations — Applied Engineering Computations

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Course	CE541E08
Department	Civil Engineering · Christ University
Instructor	Dr. Arpan Pradhan
Unit	Unit 1
Session	Day 6
CO	CO1
Topics today	Darcy-Weisbach · energy equation · pipe comparison table · complete design check

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> Run each grey code cell with **Shift + Enter**. Read the explanation above each cell first. Instructor notes are marked `—— INSTRUCTOR NOTE ——`.

```
# STUDENT HEADER — fill in before starting
student_name = "tharanaa"
roll_number  = "2460110"
github_repo  = "https://github.com/tharanaasaravanan/CE541E08-2026"
session      = "Day 6"
print("╔═══════════════════════════════════════════╗")
print("║        CE541E08 — Student Record          ║")
print("╠═══════════════════════════════════════════╣")
print(f"║  Name    :{student_name:<31}║")
print(f"║  Roll    :{roll_number:<31}║")
print(f"║  Session :{session:<31}║")
print("╚═══════════════════════════════════════════╝")
```

```
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║        CE541E08 — Student Record          ║
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║  Name    :tharanaa                        ║
║  Roll    :2460110                         ║
║  Session :Day 6                           ║
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```

Section 1 — Darcy-Weisbach head loss

$$h_f = f \cdot \frac{L}{D} \cdot \frac{V^2}{2g}$$

Translate fraction-heavy formulas carefully — use brackets.

▶ Cell 1 — Darcy-Weisbach

```
import math
L=250.0; D=0.3; V=1.8; f=0.018; g=9.81

h_f  = f * (L/D) * (V**2 / (2*g))
area = math.pi * (D/2)**2
Q    = V * area

print(f"Head loss h_f     : {h_f:.3f} m")
print(f"Discharge Q       : {Q*1000:.2f} L/s")
print(f"Hydraulic gradient: {h_f/L*1000:.2f} m/km")

# ─── INSTRUCTOR NOTE ──────────────────────────────────────────
# Hydraulic gradient = h_f / L. For water mains: 1–5 m/km typical.
# Higher value here → consider larger diameter.
# ─────────────────────────────────────────────────────────────
```

```
Head loss h_f     : 2.477 m
Discharge Q       : 127.23 L/s
Hydraulic gradient: 9.91 m/km
```

▶ Cell 2 — Compare pipe diameters for same flow

```
import math
Q=0.05; L=500.0; f=0.018; g=9.81

print(f"{'D (mm)':>8} {'V (m/s)':>10} {'h_f (m)':>10} {'grad (m/km)':>13}")
print("─" * 45)
for d_mm in [200, 250, 300, 350]:
    D    = d_mm/1000
    area = math.pi*(D/2)**2
    V    = Q/area
    h_f  = f*(L/D)*(V**2/(2*g))
    print(f"{d_mm:>8} {V:>10.3f} {h_f:>10.3f} {h_f/L*1000:>13.2f}")

# ─── INSTRUCTOR NOTE ──────────────────────────────────────────
# As diameter increases, head loss drops sharply (V⁴ effect).
# Ask: which diameter gives < 5 m/km gradient?
```

```
# Answer: 300 mm (3.99) or 350 mm (1.50).
# ──────────────────────────────────────────────────────────────
```

D (mm)	V (m/s)	h_f (m)	grad (m/km)
200	1.592	5.810	11.62
250	1.019	1.904	3.81
300	0.707	0.765	1.53
350	0.520	0.354	0.71

▶ Cell 3 — Energy equation: HGL check

```
g=9.81; gamma=9810
z1=12.5; p1=85000; V1=1.2
z2=10.2; p2=95000; V2=1.2

H1 = z1 + p1/gamma + V1**2/(2*g)
H2 = z2 + p2/gamma + V2**2/(2*g)
h_f = H1 - H2

print(f"Total head H1 = {H1:.3f} m")
print(f"Total head H2 = {H2:.3f} m")
print(f"Head loss h_f = {h_f:.3f} m")
print("(Positive h_f = energy lost in flow direction ✓)")
```

```
Total head H1 = 21.238 m
Total head H2 = 19.957 m
Head loss h_f = 1.281 m
(Positive h_f = energy lost in flow direction ✓)
```

▶ Cell 4 — Complete pipe design check (synthesis: Days 1–6)

```
import math
Q=0.08; L=350.0; D=0.35; n=0.013; f=0.018; g=9.81; nu=1.007e-6

A    = math.pi*(D/2)**2
P    = math.pi*D
R    = A/P
V    = Q/A
S    = (Q*n/(A*R**(2/3)))**2    # required slope
h_f  = f*(L/D)*(V**2/(2*g))
Re   = V*D/nu
vel_ok  = 0.6 <= V <= 3.0
turb_ok = Re > 4000

print("="*50)
print("   PIPE DESIGN — SYNTHESIS CE541E08 Day 6")
print("="*50)
```

```
    print(f"  Diameter    : {D*1000:.0f} mm")
    print(f"  Velocity    : {V:.3f} m/s  {'✓' if vel_ok else '✗'}")
    print(f"  Reynolds    : {Re:.0f}  {'Turbulent ✓' if turb_ok else '✗'}")
    print(f"  Required S  : 1 in {1/S:.0f}")
    print(f"  Head loss   : {h_f:.3f} m over {L:.0f} m")
    print(f"  STATUS      : {'ACCEPTABLE ✓' if vel_ok and turb_ok else 'REVIEW ✗'}
    print("="*50)
```

```
==================================================
   PIPE DESIGN — SYNTHESIS CE541E08 Day 6
==================================================
  Diameter    : 350 mm
  Velocity    : 0.832 m/s  ✓
  Reynolds    : 289003  Turbulent ✓
  Required S  : 1 in 332
  Head loss   : 0.634 m over 350 m
  STATUS      : ACCEPTABLE ✓
==================================================
```

Day 6 Assignment — Gravity water supply

Reservoir A: 185.0 m, Reservoir B: 162.5 m. Pipe: 1200 m, 250 mm, n=0.013, f=0.020. Compute: Δh, S=Δh/L, V (Manning's), Q, h_f (Darcy-Weisbach, verify ≈ Δh), Re. *(Δh=22.5 m, V≈2.055 m/s, Q≈100.8 L/s)*

▶ Assignment cell

```
import math
zA=185.0; zB=162.5; L=1200.0; D_mm=250; n=0.013; f=0.020; g=9.81; nu=1.007e-6
D=D_mm/1000

delta_h = zA - zB
A = math.pi * (D/2)**2
P = math.pi * D
R = A/P
S = delta_h / L

# Calculate V using Manning's equation (rearranged to solve for V)
V = (1/n) * (R**(2/3)) * (S**(1/2))

Q = V * A

h_f_DW = f * (L/D) * (V**2 / (2*g))
Re = V * D / nu

print(f"Δh={delta_h} m, S=1 in {1/S:.0f}")
print(f"V={V:.3f} m/s, Q={Q*1000:.2f} L/s")
```

```
print(f"h_f(D-W)={h_f_DW:.3f} m (should ≈ {delta_h} m)")
print(f"Re={Re:.0f}")
```

```
Δh=22.5 m, S=1 in 53
V=1.659 m/s, Q=81.43 L/s
h_f(D-W)=13.465 m (should ≈ 22.5 m)
Re=411833
```

Before next session — checklist

- [ ] Completed all assignment cells
- [ ] Saved notebook to Google Drive
- [ ] Uploaded to GitHub: `Unit1_PythonBasics/CE541E08_U1_Day6.ipynb`
- [ ] Commit message: `Day 6 assignment completed`

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